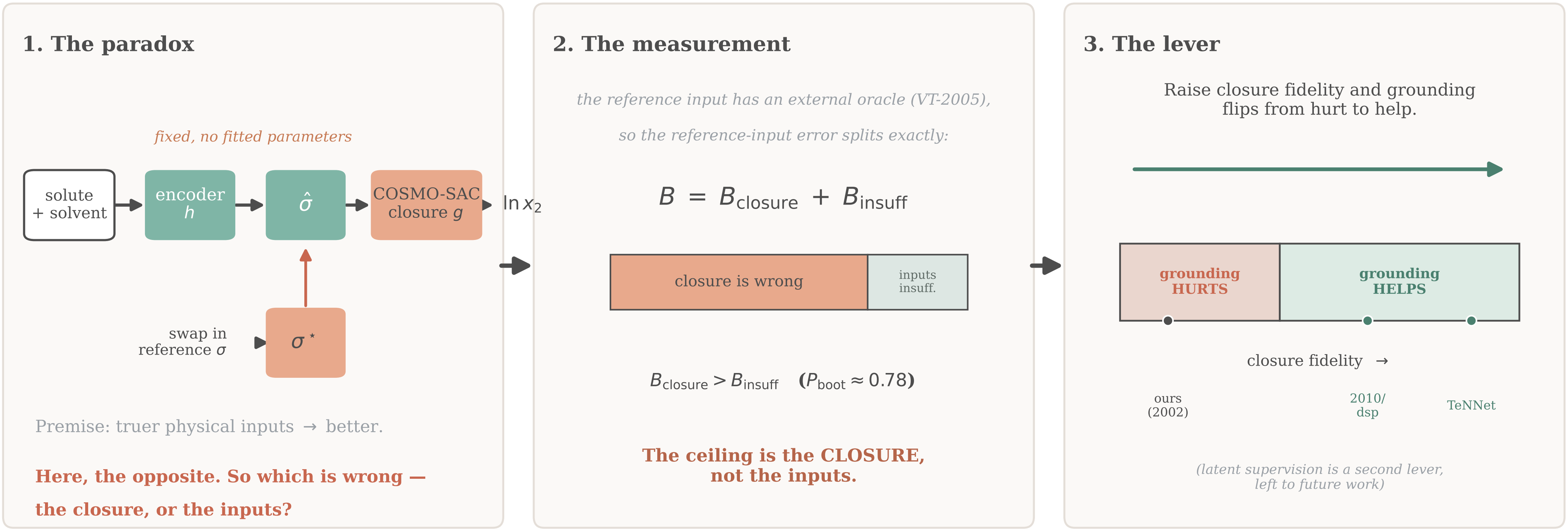
When do reference physical inputs help a learned solubility model?



What the paper establishes, graded by evidence

CERTIFIED

- Paradox: reference σ makes it worse (3 seeds, $n = 5608$)
- Exact split $B = B_{\text{closure}} + B_{\text{insuff}}$ (Lemma 2)
- Closure = NIST COSMO-SAC-2002 to $0.003 \ln\gamma$

LIKELY

- Closure binds on low-γ set: $P_{\text{boot}} \approx 0.78$
- A modern closure (2010/dsp) removes the penalty
- Latent = low-rank, transferable surrogate ($n = 44$)

EXPLORATORY

- Transfer to the finite-composition regime
- γ-stratified strengthening (estimator-dependent)
- pKa second closure as a fidelity cross-check